

MATH 142: Final Exam
Spring —₁ 2026
04/30/2026
150 Minutes

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Write your name on the appropriate line on the exam cover sheet. This exam contains 16 pages (including this cover page) and 15 questions. Check that you have every page of the exam. Answer the questions in the spaces provided on the question sheets. Be sure to answer every part of each question and show all your work. If you run out of room for an answer, continue on the back of the page — being sure to indicate the problem number.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
8	10	
9	10	
10	10	
11	10	
12	10	
13	10	
14	10	
15	10	
Total:	150	

1. Showing all your work, complete the following:

(a) (5 points) Evaluate $\int \cos^3 \theta \, d\theta$

Recall that $\sin^2 \theta + \cos^2 \theta = 1$, so $\cos^2 \theta = 1 - \sin^2 \theta$. We then have...

$$\int \cos^3 \theta \, d\theta = \int \cos^2 \theta \cdot \cos \theta \, d\theta = \int (1 - \sin^2 \theta) \cos \theta \, d\theta$$

Now let $u = \sin \theta$, so that $du = \cos \theta \, d\theta$. But then...

$$\int \cos^3 \theta \, d\theta = \int (1 - \sin^2 \theta) \cos \theta \, d\theta = \int (1 - u^2) \, du = u - \frac{u^3}{3} + C = \sin \theta - \frac{\sin^3 \theta}{3} + C$$

Therefore,

$$\boxed{\int \cos^3 \theta \, d\theta = \sin \theta - \frac{\sin^3 \theta}{3} + C}$$

(b) (5 points) Set-up but do not evaluate an integral which computes the arclength of $y = \frac{2}{7} x^{7/2}$ on $[0, 1]$.

The arclength of a differentiable function $f(x)$ on $[a, b]$ is $L = \int_a^b \sqrt{1 + (f'(x))^2} \, dx$.

We have...

$$\begin{aligned} f(x) &= \frac{2}{7} x^{7/2} \\ f'(x) &= x^{5/2} \\ (f'(x))^2 &= (x^{5/2})^2 = x^5 \end{aligned}$$

We also have $[a, b] = [0, 1]$. Therefore, the arclength is given by...

$$\boxed{L = \int_0^1 \sqrt{1 + x^5} \, dx}$$

Note. $L = \int_0^1 \sqrt{1 + x^5} \, dx = {}_2F_1(-\frac{1}{2}, \frac{1}{5}, \frac{6}{5}, -1) \approx 1.07467$, where ${}_2F_1$ is the hypergeometric function.

2. (10 points) Evaluate the integral below.

$$\int \frac{1}{(\sqrt{16-x^2})^3} dx$$

Because of the “Pythagorean Theorem-like” term $16 - x^2$, we use Trig. Substitution. From the Pythagorean Theorem, we know that $a^2 + b^2 = c^2$. But then $b^2 = c^2 - a^2$. Aligning corresponding terms, we want $c^2 - a^2 = 16 - x^2$. So, we take $a^2 = x^2$, i.e. $a = x$, and $c^2 = 16$, i.e. $c = 4$. But then $b^2 = c^2 - a^2 = 16 - x^2$, so $b = \sqrt{16 - x^2}$. There are two possible right triangles corresponding to these sides we could draw:



Using the triangle on the left, we see that $\sin \theta = \frac{x}{4}$, i.e. $x = 4 \sin \theta$. But then $dx = 4 \cos \theta d\theta$. We can also see from the triangle that $\cos \theta = \frac{\sqrt{16-x^2}}{4}$, so that $\sqrt{16-x^2} = 4 \cos \theta$. But then $(\sqrt{16-x^2})^3 = (4 \cos \theta)^3 = 4^3 \cos^3 \theta$. Therefore, we have...

$$\int \frac{1}{(\sqrt{16-x^2})^3} dx = \int \frac{1}{4^3 \cos^3 \theta} \cdot 4 \cos \theta d\theta = \int \frac{1}{4^2 \cos^2 \theta} \cdot 4 \cos \theta d\theta = \frac{1}{16} \int \frac{1}{\cos^2 \theta} d\theta$$

Using the fact that $\cos \theta = \frac{1}{\sec \theta}$, we know $\sec \theta = \frac{1}{\cos \theta}$ and $\sec^2 \theta = \frac{1}{\cos^2 \theta}$. But then...

$$\int \frac{1}{(\sqrt{16-x^2})^3} dx = \frac{1}{16} \int \frac{1}{\cos^2 \theta} d\theta = \frac{1}{16} \int \sec^2 \theta d\theta = \frac{1}{16} \tan \theta + C$$

From the right triangle, we see that $\tan \theta = \frac{x}{\sqrt{16-x^2}}$. Therefore, we have...

$$\int \frac{1}{(\sqrt{16-x^2})^3} dx = \frac{1}{16} \tan \theta + C = \frac{1}{16} \cdot \frac{x}{\sqrt{16-x^2}} + C = \boxed{\frac{x}{16\sqrt{16-x^2}} + C}$$

Alternatively, using the triangle on the right, we see that $\cos \theta = \frac{x}{4}$, i.e. $x = 4 \cos \theta$. But then $dx = -4 \sin \theta d\theta$. We can also see from the triangle that $\sin \theta = \frac{\sqrt{16-x^2}}{4}$, so that $\sqrt{16-x^2} = 4 \sin \theta$. But then $(\sqrt{16-x^2})^3 = (4 \sin \theta)^3 = 4^3 \sin^3 \theta$. Therefore, we have...

$$\int \frac{1}{(\sqrt{16-x^2})^3} dx = \int \frac{1}{4^3 \sin^3 \theta} \cdot -4 \sin \theta d\theta = \int \frac{1}{4^2 \sin^2 \theta} \cdot -4 \sin \theta d\theta = -\frac{1}{16} \int \frac{1}{\sin^2 \theta} d\theta$$

Using the fact that $\csc \theta = \frac{1}{\sin \theta}$, we know $\csc^2 \theta = \frac{1}{\sin^2 \theta}$. But then...

$$\int \frac{1}{(\sqrt{16-x^2})^3} dx = -\frac{1}{16} \int \frac{1}{\sin^2 \theta} d\theta = \frac{1}{16} \int -\csc^2 \theta d\theta = \frac{1}{16} \cot \theta + C$$

From the right triangle, we see that $\cot \theta = \frac{x}{\sqrt{16-x^2}}$. Therefore, we have...

$$\int \frac{1}{(\sqrt{16-x^2})^3} dx = \frac{1}{16} \cot \theta + C = \frac{1}{16} \cdot \frac{x}{\sqrt{16-x^2}} + C = \boxed{\frac{x}{16\sqrt{16-x^2}} + C}$$

3. (10 points) Compute the following integral:

$$\int \frac{13x - 8}{(x - 2)(x + 1)} dx$$

We use partial fractions. The degree of the numerator is 1 and the degree of the denominator is 2, so no long division is necessary. Observe the denominator is already fully factored. So, we can apply partial fractions decomposition. We have...

$$\frac{13x - 8}{(x - 2)(x + 1)} = \frac{A}{x - 2} + \frac{B}{x + 1}$$

Combining terms on the right, we have...

$$\frac{A}{x - 2} + \frac{B}{x + 1} = \frac{A(x + 1) + B(x - 2)}{(x - 2)(x + 1)} = \frac{Ax + A + Bx - 2B}{(x - 2)(x + 1)} = \frac{(A + B)x + (A - 2B)}{(x - 2)(x + 1)}$$

Because this must be equal to $\frac{13x - 8}{(x - 2)(x + 1)}$ and the two have the same denominator, their numerators must be equal. Equating like-terms in the numerators, we obtain the system of equations...

$$\begin{cases} x: & A + B = 13 \\ 1: & A - 2B = -8 \end{cases}$$

Subtracting the second equation from the first (so the A 's cancel), we have $3B = 21$, but then $B = 7$. From the first equation, we would then have $13 = A + B = A + 7$, so $A = 6$. Alternatively, we could use Heaviside's Method (the Cover-Up Method). We would have...

$$A = \frac{13x - 8}{\boxed{(x - 2)}(x + 1)} \Big|_{x=2} = \frac{13(2) - 8}{2 + 1} = \frac{26 - 8}{3} = \frac{18}{3} = 6$$

$$B = \frac{13x - 8}{(x - 2)\boxed{(x + 1)}} \Big|_{x=-1} = \frac{13(-1) - 8}{-1 - 2} = \frac{-13 - 8}{-3} = \frac{-21}{-3} = 7$$

Therefore, we have...

$$\int \frac{13x - 8}{(x - 2)(x + 1)} dx = \int \left(\frac{6}{x - 2} + \frac{7}{x + 1} \right) dx = 6 \ln |x - 2| + 7 \ln |x + 1| + K$$

Therefore, we have...

$$\boxed{\int \frac{13x - 8}{(x - 2)(x + 1)} dx = 6 \ln |x - 2| + 7 \ln |x + 1| + K}$$

4. (10 points) Showing all your work, compute the integral below.

$$\int_0^{\pi} x \sin(7x) \, dx$$

We use *Integration-by-Parts*. Using *LIATE*, we choose $u = x$ (*A* for algebraic) and $dv = \sin(7x)$, i.e.

x	$-\frac{\cos(7x)}{7}$
1	$\sin(7x)$

So, we have...

$$\begin{aligned}
 \int_0^{\pi} x \sin(7x) \, dx &= -\frac{x \cos(7x)}{7} \Big|_0^{\pi} - \int_0^{\pi} -\frac{\cos(7x)}{7} \, dx \\
 &= \left(-\frac{\pi \cos(7\pi)}{7} - \frac{-0 \cos(0)}{7} \right) + \frac{1}{7} \int_0^{\pi} \cos(7x) \, dx \\
 &= -\frac{\pi(-1)}{7} + \frac{1}{7} \cdot \frac{\sin(7x)}{7} \Big|_0^{\pi} \\
 &= \frac{\pi}{7} + \frac{1}{49} (\sin(7\pi) - \sin(0)) \\
 &= \frac{\pi}{7} + \frac{1}{49} (0 - 0) \\
 &= \boxed{\frac{\pi}{7}}
 \end{aligned}$$

Alternatively, we can evaluate this integral using *tabular integration*:

u	dv
x	$\sin(7x)$
1	$-\frac{\cos(7x)}{7}$
0	$-\frac{\sin(7x)}{49}$

So that...

$$\begin{aligned}
 \int_0^{\pi} x \sin(7x) \, dx &= \left(-\frac{x \cos(7x)}{7} + \frac{\sin(7x)}{49} \right) \Big|_0^{\pi} \\
 &= \left(-\frac{\pi \cos(7\pi)}{7} + \frac{\sin(7\pi)}{49} \right) - \left(-\frac{0 \cos(0)}{7} + \frac{\sin(0)}{49} \right) \\
 &= \left(-\frac{\pi(-1)}{7} + \frac{\sin(0)}{49} \right) - (0 + 0) \\
 &= \boxed{\frac{\pi}{7}}
 \end{aligned}$$

5. (10 points) Determine whether the following improper integral converges or diverges. Be sure to fully justify your response.

$$\int_0^{\infty} \frac{dx}{(x+4)^2}$$

We have...

$$\int \frac{dx}{(x+4)^2} = -\frac{1}{x+4} + C$$

Therefore, we have...

$$\begin{aligned} \int_0^{\infty} \frac{dx}{(x+4)^2} &:= \lim_{b \rightarrow \infty} \int_0^b \frac{dx}{(x+4)^2} \\ &= \lim_{b \rightarrow \infty} \left. -\frac{1}{x+4} \right|_0^b \\ &= \lim_{b \rightarrow \infty} -\frac{1}{b+4} - \left(-\frac{1}{0+4} \right) \\ &= 0 - \left(-\frac{1}{4} \right) \\ &= \frac{1}{4} \end{aligned}$$

Therefore, the integral converges to $\frac{1}{4}$.

$$\boxed{\int_0^{\infty} \frac{dx}{(x+4)^2} = \frac{1}{4}}$$

6. Determine whether the following series converge or diverge. Be sure to fully justify your response.

(a) (5 points) $\sum_{n=1}^{\infty} \frac{3n+5}{7n-4}$

Observe...

$$\lim_{n \rightarrow \infty} \frac{3n+5}{7n-4} = \frac{3}{7} \neq 0$$

Therefore, $\sum_{n=1}^{\infty} \frac{3n+5}{7n-4}$ diverges by the Divergence Test.

(b) (5 points) $\sum_{n=0}^{\infty} \left(\frac{1}{2^k} - \frac{1}{2^{k+1}} \right)$

Observe that...

$$\begin{aligned} \sum_{n=0}^{\infty} \left(\frac{1}{2^k} - \frac{1}{2^{k+1}} \right) &= \lim_{N \rightarrow \infty} \sum_{n=0}^N \left(\frac{1}{2^k} - \frac{1}{2^{k+1}} \right) \\ &= \lim_{N \rightarrow \infty} \left[\left(\frac{1}{2^0} - \frac{1}{2^1} \right) + \left(\frac{1}{2^1} - \frac{1}{2^2} \right) + \left(\frac{1}{2^2} - \frac{1}{2^3} \right) + \cdots + \left(\frac{1}{2^N} - \frac{1}{2^{N+1}} \right) \right] \\ &= \lim_{N \rightarrow \infty} \left[1 - \frac{1}{2^{N+1}} \right] \\ &= 1 \end{aligned}$$

OR

$$\sum_{n=0}^{\infty} \left(\frac{1}{2^k} - \frac{1}{2^{k+1}} \right) = \sum_{n=0}^{\infty} \left(\frac{2}{2^{k+1}} - \frac{1}{2^{k+1}} \right) = \sum_{n=0}^{\infty} \left(\frac{2-1}{2^{k+1}} \right) = \sum_{n=0}^{\infty} \left(\frac{1}{2^{k+1}} \right) = \sum_{n=0}^{\infty} \frac{1}{2} \left(\frac{1}{2} \right)^k$$

This series is geometric with $|r| = \frac{1}{2} < 1$. Therefore, the series converges by the

Geometric Series Test. In fact, the sum would be $\frac{\frac{1}{2}}{1 - \frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$.

7. (10 points) Determine whether the following series converges or diverges. Be sure to fully justify your response. If the series converges, find its sum.

$$\sum_{n=2}^{\infty} 3 \left(-\frac{2}{5} \right)^n$$

The series is geometric because the series has the form $\sum_{n=2}^{\infty} ar^n$ with $a = 3$ and $r = -\frac{2}{5}$. Because $|r| = \frac{2}{5} < 1$, the series converges by the Geometric Series Test. We know that if a geometric series converges, it converges to $\frac{\text{first term}}{1-r}$. Therefore, we have...

$$\frac{\text{first term}}{1-r} = \frac{3 \left(-\frac{2}{5} \right)^2}{1 - \left(-\frac{2}{5} \right)} = \frac{3 \cdot \frac{4}{25}}{1 + \frac{2}{5}} \cdot \frac{25}{25} = \frac{3 \cdot 4}{25 + 2(5)} = \frac{12}{25 + 10} = \frac{12}{35} \approx 0.342857$$

$$\boxed{\sum_{n=2}^{\infty} 3 \left(-\frac{2}{5} \right)^n \text{ converges by G.S.T. to } \frac{12}{35}}$$

8. (10 points) Being sure to fully support your reasoning, determine whether the series given below converges or diverges.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{7+2n}$$

Observe that the series is an alternating series. We have...

- $\lim_{n \rightarrow \infty} \frac{1}{7+2n} = 0$: This function is rational. The degree of the denominator is 1 while the degree of the numerator is 0. Therefore, the limit at $\pm\infty$ is 0.
- $\left\{ \frac{1}{7+2n} \right\}$ is decreasing: We need to show that $\{a_n\}$ is decreasing, i.e. $a_{n+1} < a_n$. It is clear that $9+2n > 7+2n$, so that $\frac{1}{9+2n} < \frac{1}{7+2n}$. But $\frac{1}{9+2n} = \frac{1}{7+2+2n} = \frac{1}{7+2(n+1)}$. But then $a_{n+1} = \frac{1}{7+2(n+1)} = \frac{1}{9+2n} < \frac{1}{7+2n} = a_n$, so the sequence is decreasing. Alternatively, $\frac{d}{dx} \left(\frac{1}{7+2x} \right) = -\frac{2}{(7+2x)^2} < 0$ for all x . So, the sequence is decreasing.

Therefore, the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{7+2n}$ converges by the Alternating Series Test.

$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{7+2n} \text{ converges by A.S.T.}$

9. Determine whether the following series converge or diverge. Be sure to fully justify your response.

(a) (5 points) $\sum_{n=0}^{\infty} \frac{(-3)^n}{n!}$

Using the Ratio Test, we have...

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{(-3)^{n+1}}{(n+1)!}}{\frac{(-3)^n}{n!}} \right| = \lim_{n \rightarrow \infty} \frac{3^{n+1}}{(n+1)!} \cdot \frac{n!}{3^n} = \lim_{n \rightarrow \infty} \frac{3^{n+1}}{3^n} \cdot \frac{n!}{(n+1)!} = \lim_{n \rightarrow \infty} \frac{3^n \cdot 3}{3^n} \cdot \frac{n!}{(n+1)n!} = \lim_{n \rightarrow \infty} \frac{3}{n+1} = 0 < 1$$

Therefore, the series converges absolutely by the Ratio Test.

(b) (5 points) $\sum_{n=1}^{\infty} \left(\frac{3n+4}{2n} \right)^n$

Using the Root Test, we have...

$$\lim_{n \rightarrow \infty} \left| \left(\frac{3n+4}{2n} \right)^n \right|^{1/n} = \lim_{n \rightarrow \infty} \frac{3n+4}{2n} = \frac{3}{2} > 1$$

Therefore, the series diverges by the Root Test.

10. (10 points) Determine whether the series below diverges, converges conditionally, or converges absolutely. Include all necessary justifications. [Hint: $n > \ln(n+1)$ for $n \geq 1$.]

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\ln(n+1)}$$

Observe that the given series is alternating. We can also observe...

- $\lim_{n \rightarrow \infty} \frac{1}{\ln(n+1)} = 0$: This is obvious as the numerator is fixed but $\lim_{n \rightarrow \infty} \ln(n+1) = \infty$.
- $\left\{ \frac{1}{\ln(n+1)} \right\}$ is decreasing: We need to show that $a_{n+1} < a_n$. Observe that $n+1 > n$. As $\ln x$ is an increasing function, observe $\ln(n+1) > \ln(n)$. But then $a_{n+1} = \frac{1}{\ln(n+1)} < \frac{1}{\ln n} = a_n$. So, the sequence is decreasing. Alternatively, observe that $\frac{d}{dx} \left(\frac{1}{\ln(x+1)} \right) = -\frac{1}{(x+1)[\ln(x+1)]^2} < 0$ for $x > -1$. This also shows the sequence is decreasing.

Therefore, the given series converges by the Alternating Series Test.

Now we consider the series $\sum_{n=1}^{\infty} \frac{1}{\ln(n+1)}$. We know that $n > \ln(n+1)$ for $n \geq 1$. But then $\frac{1}{n} < \frac{1}{\ln(n+1)}$ for $n \geq 1$. But then we know...

$$\sum_{n=1}^{\infty} \frac{1}{n} < \sum_{n=1}^{\infty} \frac{1}{\ln(n+1)}$$

Because the series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges by the p -test with $p = 1$. Alternatively, this is the

Harmonic series, which we know diverges. But then the series $\sum_{n=1}^{\infty} \frac{1}{\ln(n+1)}$ diverges by the Direct Comparison Test.

Because $\sum_{n=1}^{\infty} \frac{(-1)^n}{\ln(n+1)}$ converges but the series $\sum_{n=1}^{\infty} \frac{1}{\ln(n+1)}$ diverges, the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\ln(n+1)}$ converges conditionally.

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\ln(n+1)} \text{ converges conditionally}$$

11. Determine the center, radius, and interval of convergence for the following power series:

(a) (5 points) $\sum_{n=0}^{\infty} n! (x + 5)^n$

Using the Ratio Test, we have...

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)!(x+5)^{n+1}}{n!(x+5)^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)n!(x+5)^n(x+5)}{n!(x+5)^n} \right| = \lim_{n \rightarrow \infty} |(n+1)(x+5)| = \infty$$

Unless, of course, $x = -5$ (the center) in which case the series converges (it is 0). So, the center is $x = -5$, the radius of convergence is $R = 0$, and the “interval” of convergence is $\{5\}$.

Center: $x = -5$
“Interval” of Convergence: $\{5\}$
Radius of Convergence: $R = 0$

(b) (5 points) $\sum_{n=0}^{\infty} \frac{x^n}{e^{n^2}}$ [Hint: Use the Root Test]

The center is $x = 0$. Using the Root Test, we have...

$$\lim_{n \rightarrow \infty} \left| \frac{x^n}{e^{n^2}} \right|^{1/n} = \lim_{n \rightarrow \infty} \left| \frac{x}{e^n} \right| = 0$$

Therefore, the series converges for all x . So, the interval of convergence is $(-\infty, \infty)$, which means the radius of convergence is $R = \infty$.

Center: $x = 0$
Interval of Convergence: $(-\infty, \infty)$
Radius of Convergence: $R = \infty$

12. (10 points) Find the center, radius, and interval of convergence for the following power series:

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{2^n \sqrt{n}} (x-1)^n$$

Clearly, the center is $x = 1$. We find the interval of convergence. Using the Ratio Test, we have...

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{\frac{(-1)^{n+1}}{2^{n+1} \sqrt{n+1}} (x-1)^{n+1}}{\frac{(-1)^n}{2^n \sqrt{n}} (x-1)^n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(x-1)^{n+1}}{2^{n+1} \sqrt{n+1}} \cdot \frac{2^n \sqrt{n}}{(x-1)^n} \right| \\ &= \left| \frac{(x-1)^{n+1}}{(x-1)^n} \cdot \frac{2^n}{2^{n+1}} \cdot \frac{\sqrt{n}}{\sqrt{n+1}} \right| \\ &= \left| (x-1) \cdot \frac{1}{2} \cdot \sqrt{\frac{n}{n+1}} \right| \\ &= \left| \frac{x-1}{2} \right| \end{aligned}$$

Alternatively, using the Root Test, we have...

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^n}{2^n \sqrt{n}} (x-1)^n \right|^{1/n} = \lim_{n \rightarrow \infty} \left| \frac{1}{2n^{1/(2n)}} (x-1) \right| = \lim_{n \rightarrow \infty} \left| \frac{1}{2(n^{1/n})^2} (x-1) \right| = \left| \frac{1}{2(1)^2} (x-1) \right| = \left| \frac{x-1}{2} \right|$$

In either case, we know the series will converge if $\left| \frac{x-1}{2} \right| < 1$, i.e. $-1 < \frac{x-1}{2} < 1$.

But then $-2 < x-1 < 2$, so that $-1 < x < 3$. Thus, the radius of convergence is $R = \frac{3-(-1)}{2} = \frac{3+1}{2} = \frac{4}{2} = 2$. We now need to test the endpoints.

$x = -1$: If $x = -1$, the series is $\sum_{n=1}^{\infty} \frac{(-1)^n}{2^n \sqrt{n}} (-1-1)^n = \sum_{n=1}^{\infty} \frac{(-1)^n}{2^n \sqrt{n}} (-2)^n = \sum_{n=1}^{\infty} \frac{(-1)^n (-1)^n}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$. This series diverges by the p -test with $p = \frac{1}{2}$.

$x = 3$: If $x = 3$, the series is $\sum_{n=1}^{\infty} \frac{(-1)^n}{2^n \sqrt{n}} (3-1)^n = \sum_{n=1}^{\infty} \frac{(-1)^n}{2^n \sqrt{n}} 2^n = \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$. This series is alternating. We see $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$ and the sequence $\left\{ \frac{1}{\sqrt{n}} \right\}$ is decreasing. Therefore, this series converges by the Alternating Series Test.

Center: $x = 1$
Interval of Convergence: $(-1, 3]$
Radius of Convergence: $R = 2$

13. (10 points) Find the third-degree Taylor polynomial, $T_3(x)$, for $f(x) = \sqrt{x}$ centered at $x = 1$.

We know that the Taylor series for a function $f(x)$ centered at $x = c$ is...

$$T(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x - c)^n$$

The center here is $c = 1$. We also want the third-degree Taylor polynomial, i.e. the terms of the Taylor series up to $n = 3$. Therefore, we have...

$$\begin{aligned} T_3(x) &= \sum_{n=0}^3 \frac{f^{(n)}(1)}{n!} (x - 1)^n \\ &= \frac{f^{(0)}(1)}{0!} (x - 1)^0 + \frac{f^{(1)}(1)}{1!} (x - 1)^1 + \frac{f^{(2)}(1)}{2!} (x - 1)^2 + \frac{f^{(3)}(1)}{3!} (x - 1)^3 \\ &= f(1) + f'(1)(x - 1) + \frac{f''(1)}{2} (x - 1)^2 + \frac{f'''(1)}{6} (x - 1)^3 \end{aligned}$$

We need only find these derivative values. We have...

$$\begin{aligned} f(x) &= \sqrt{x} \Big|_{x=1} = \sqrt{1} = 1 \\ f'(x) &= \frac{1}{2\sqrt{x}} \Big|_{x=1} = \frac{1}{2\sqrt{1}} = \frac{1}{2} \\ f''(x) &= -\frac{1}{4x^{3/2}} \Big|_{x=1} = -\frac{1}{4(1^{3/2})} = -\frac{1}{4} \\ f'''(x) &= \frac{3}{8x^{5/2}} \Big|_{x=1} = \frac{3}{8(1^{5/2})} = \frac{3}{8} \end{aligned}$$

Therefore, we have...

$$\begin{aligned} T_3(x) &= f(1) + f'(1)(x - 1) + \frac{f''(1)}{2} (x - 1)^2 + \frac{f'''(1)}{6} (x - 1)^3 \\ &= 1 + \frac{1}{2} (x - 1) + \frac{-1/4}{2} (x - 1)^2 + \frac{3/8}{6} (x - 1)^3 \\ &= 1 + \frac{1}{2} (x - 1) - \frac{1}{8} (x - 1)^2 + \frac{1}{16} (x - 1)^3 \end{aligned}$$

Therefore, we have...

$$\boxed{T_3(x) = 1 + \frac{1}{2} (x - 1) - \frac{1}{8} (x - 1)^2 + \frac{1}{16} (x - 1)^3}$$

14. Use a known Maclaurin series to find a...

(a) (5 points) ... power series representation for $\frac{1}{1+x^2}$.

We know the Maclaurin series for $\frac{1}{1-x}$ is $\sum_{n=0}^{\infty} x^n$. But then the Maclaurin series for $\frac{1}{1+x^2}$ is...

$$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty} (-x^2)^n = \boxed{\sum_{n=0}^{\infty} (-1)^n x^{2n}}$$

(b) (5 points) ... series approximation to $\int_0^1 \ln(1+x^2) dx$.

We know the Maclaurin series for $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$ for $x \in (-1, 1]$. Observe that $x \in [0, 1]$, so that $x^2 \in [0, 1]$. But then...

$$\ln(1+x^2) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(x^2)^n}{n} = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^{2n}}{n}$$

But then...

$$\begin{aligned} \int_0^1 \ln(1+x^2) dx &= \int_0^1 \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^{2n}}{n} dx \\ &= \sum_{n=1}^{\infty} \int_0^1 (-1)^{n+1} \frac{x^{2n}}{n} dx \\ &= \sum_{n=1}^{\infty} (-1)^{n+1} \left. \frac{x^{2n+1}}{n(2n+1)} \right|_0^1 \\ &= \sum_{n=1}^{\infty} \left((-1)^{n+1} \frac{1^{2n+1}}{n(2n+1)} - (-1)^{n+1} \frac{0^{2n+1}}{n(2n+1)} \right) \\ &= \boxed{\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n(2n+1)}} = \frac{\ln(4) + \pi - 4}{2} \approx 0.263944 \end{aligned}$$

15. (10 points) Determine the sum of the following series:

(a) $12 - 6 + 3 - \frac{3}{2} + \frac{3}{4} - \frac{3}{8} + \dots$

Observe that each term in the sum/difference is $-\frac{1}{2}$ times the previous. Therefore, this series is geometric with common ratio $r = -\frac{1}{2}$ and first term 12. Because $|r| = \frac{1}{2} < 1$, the sum converges. We know the sum is then $\frac{\text{first term}}{1-r} = \frac{12}{1-(-\frac{1}{2})} = \frac{12}{1+\frac{1}{2}} = \frac{12}{\frac{3}{2}} = \frac{12}{1} \cdot \frac{2}{3} = \frac{24}{3} = \boxed{8}$.

Alternatively, we know that $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $x \in (-1, 1)$. But then $\frac{12}{1-x} = \sum_{n=0}^{\infty} 12x^n$ for $x \in (-1, 1)$. Observe the given series is this series with $x = -\frac{1}{2} \in (-1, 1)$. But then...

$$12 - 6 + 3 - \frac{3}{2} + \frac{3}{4} - \frac{3}{8} + \dots = \sum_{n=0}^{\infty} 12 \left(-\frac{1}{2}\right)^n = \sum_{n=0}^{\infty} 12x^n \Big|_{x=-\frac{1}{2}} = \frac{12}{1-x} \Big|_{x=-\frac{1}{2}} = \frac{12}{1-(-\frac{1}{2})} = \frac{12}{1+\frac{1}{2}} = \frac{12}{\frac{3}{2}} = \frac{12}{1} \cdot \frac{2}{3} = \frac{24}{3} = \boxed{8}$$

(b) $\sum_{n=0}^{\infty} (-1)^n \frac{(0.3)^{2n+1}}{(2n+1)!}$

We know that $\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$ for all $x \in (-\infty, \infty)$. We know $x = 0.3 \in (-\infty, \infty)$. But then...

$$\sum_{n=0}^{\infty} (-1)^n \frac{(0.3)^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \Big|_{x=0.3} = \sin(x) \Big|_{x=0.3} = \boxed{\sin(0.3)} \approx 0.29552$$

Note. We would anticipate the value of the series to be approximately 0.3 because we know $\sin \theta \approx \theta$ when $\theta \approx 0$.

(c) $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$

We know that $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$ for $x \in (-1, 1]$. But $x = 1 \in (-1, 1]$, so that...

$$\begin{aligned} 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots &= \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} \\ &= \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \Big|_{x=1} \\ &= \ln(1+x) \Big|_{x=1} \\ &= \ln(1+1) \\ &= \boxed{\ln(2)} \approx 0.693147 \end{aligned}$$